DynaCampus-FR

Complete Project Context and Execution Blueprint

Domain-Adaptive, Quality-Aware Open-Set Face Recognition with Training-Free Dynamic Enrollment for Low-Quality Campus CCTV

A self-contained context document for researchers, supervisors, developers, annotators, and AI/LLM assistants

Version 1.0 | July 2026

Department of Computer Science and Engineering
United International University

Document status: SOURCE OF TRUTH FOR THE REVISED PROJECT

Document Control and How to Use This File

Purpose of this document
This file contains the complete context, fixed design choices, research rationale, datasets, methods, implementation plan, experiment protocol, ethical safeguards, folder structures, commands, milestones, and paper strategy for the project. A person or AI assistant should be able to understand the project without receiving any additional background.

Field	Value
Project short name	DynaCampus-FR
Primary title	Domain-Adaptive, Quality-Aware Open-Set Face Recognition with Training-Free Dynamic Enrollment for Low-Quality Campus CCTV
Institution	United International University, Department of CSE
Project type	Four-month, research-first undergraduate final-year design project
Starting condition	Beginner team; no usable dataset, no validated recognition pipeline, and no finalized research protocol
Primary audience	Research team, supervisor, external reviewer, developer, annotator, or AI/LLM assistant
Authority	This file supersedes earlier app-centered plans when they conflict with the revised research direction
Publication status	The project is designed for publishable-quality research; publication is not guaranteed and depends on experimental rigor and results

Instructions for an AI or New Team Member

1. Read the Executive Summary, Fixed Decisions, Research Questions, Data Protocol, and Experiment Protocol before proposing changes.
2. Treat the stated core path as fixed unless the supervisor explicitly approves a change. Do not return to a closed-set student classifier or an app-first attendance system.
3. Do not assume that any dataset has already been collected. The project begins with zero local data and requires permission, consent, collection, verification, and identity-disjoint splits.
4. Do not claim novelty merely from using a newer model. The intended contribution is the combination of dynamic enrollment, cross-domain adaptation, CCTV-aware enrollment, quality-aware temporal aggregation, and calibrated open-set evaluation.
5. Keep all biometric data private. Code may be shared; raw identifiable face data must not be uploaded to a public repository.
6. When generating code, use the repository structure, configuration conventions, data schemas, and stage gates defined in this document.

Team Context

The source FYDP report lists a five-member team: Md. Raisul Islam, Rajkonna-Al Farabi, Alvi Islam, Shamsiya Nasrin Chyti, and Shahedul Islam Shahed. Student identification numbers are intentionally omitted from this general handoff document. The original report was submitted on July 11, 2026. [P0]

Contents at a Glance

Part	Coverage
I. Project Definition	Original baseline, problem diagnosis, final singular direction, objectives, hypotheses, scope, terminology
II. Method and Architecture	Detection, alignment, embeddings, adapter, dynamic enrollment, tracking, quality, aggregation, open-set decision, ViT comparison
III. Dataset and Ethics	Public datasets, CampusFace-BD, consent, collection, annotation, metadata, splits, leakage prevention
IV. Technical Implementation	Platforms, tech stack, hardware, repository, environment, scripts, configuration, coding workflow
V. Experiments and Evaluation	Ablations, metrics, calibration, statistics, efficiency, result tables, failure analysis
VI. Four-Month Execution	Week-by-week roadmap, first seven days, team roles, milestone gates, risk controls
VII. Paper and Handoff	Paper structure, claims, limitations, reproducibility package, demo, LLM startup prompt, references

Executive Summary

The existing project is a CCTV-based face detection and recognition system for educational institutions. Its implemented baseline uses LFW images from seven identities, 40 x 40 face crops, Haar Cascade detection, edge-based preprocessing, PCA-KNN, a small seven-class CNN, an 85% softmax confidence threshold for unknown rejection, and desktop/dashboard software. The report records 100% training accuracy, approximately 82-86% validation performance, and complete rejection of 400 Olivetti images. These results demonstrate a prototype, but not a strong open-set surveillance recognition study. [P0]

The revised project will not train a new output class for every student. It will use a fixed pretrained embedding model. New students and faculty will be enrolled by adding one quality-controlled identity template to a gallery. The research will address the real domain gap between clean enrollment photographs and low-quality CCTV probes, and it will make decisions using multiple frames from a tracked video sequence rather than a single image.

The final core method is: SCRFD face detection, five-point face alignment, AdaFace embeddings, a lightweight residual CCTV Domain Adapter trained only on development identities, CCTV-aware augmented enrollment, quality-aware temporal aggregation, exact cosine gallery matching, and a calibrated four-state open-set decision. ArcFace is the baseline. FaceLiVTv2-S is a secondary exploratory CNN-Transformer comparison that is added only after the primary AdaFace experiments are complete. [R1-R5, R10]

The project will create CampusFace-BD, a consent-based local research dataset with controlled enrollment media, two CCTV-style sessions on different days, two camera viewpoints, multiple distances, lighting and occlusion conditions, and genuine unknown identities. Public evaluation will use QMUL-SurvFace as the main surveillance benchmark when access is available, with SCface and RFW as supporting evaluations and WIDER FACE only for detector context. [R6-R9]

One-sentence project definition

DynaCampus-FR studies whether training-free dynamic enrollment, CCTV-aware cross-domain adaptation, and quality-aware multi-frame aggregation can improve open-set campus face identification under low-resolution surveillance conditions without retraining the recognition backbone.

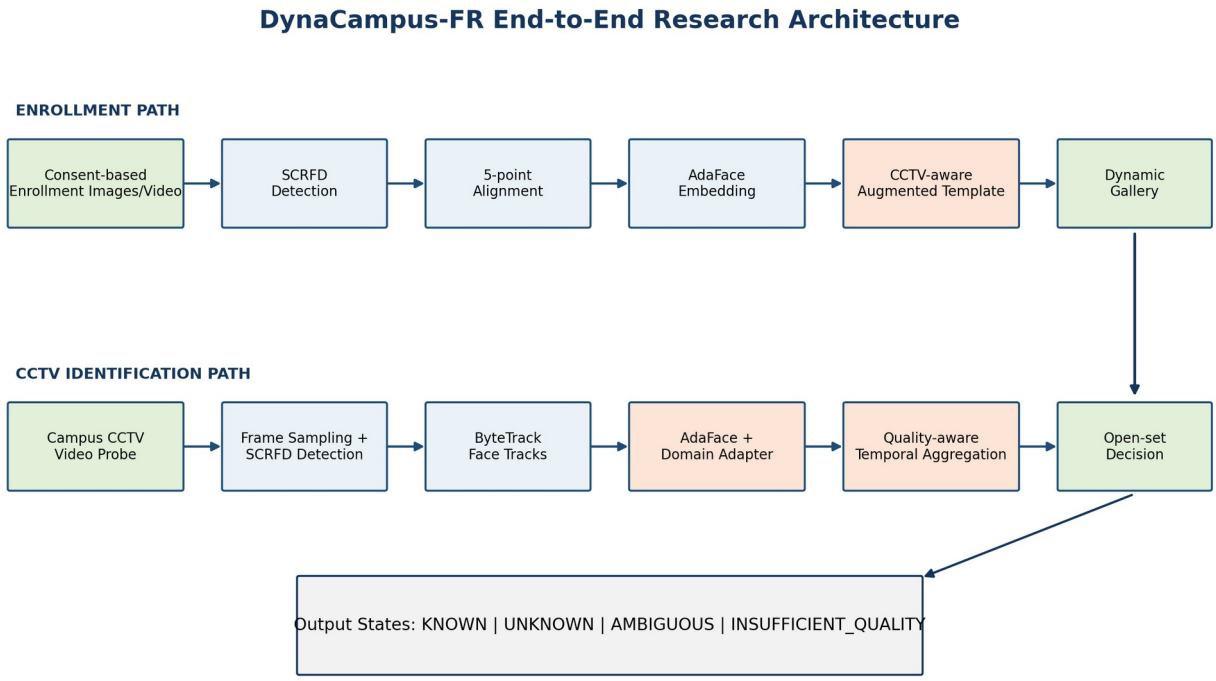


Figure 1. Final end-to-end research architecture.

PART I

PROJECT DEFINITION

What the project is, why the old direction is insufficient, and what is fixed now

1. Original Project Baseline and Starting Point

1.1 Original title and intention

The source report is titled “A Real-Time CCTV-Based Face Detection and Recognition System for Educational Institutions Using Lightweight Deep Learning Models.” It motivates automatic student verification under blur, poor lighting, occlusion, non-frontal pose, low resolution, and limited hardware. It also discusses privacy, real-time speed, and model comparison. [P0]

1.2 Existing implemented pipeline

Component	Current report implementation	Research limitation
Dataset	Filtered LFW: 1,288 images, seven identities, 80:20 split	Not a campus CCTV dataset; extremely small identity count; image-level split may not represent surveillance generalization
Input	40 x 40 face images	Too small for modern pretrained face encoders and loses identity detail
Preprocessing	Gaussian filtering, Sobel edges, Canny edges	Not aligned with modern embedding models; can remove identity texture
Detection	Viola-Jones Haar Cascade	Outdated and weak for small, rotated, blurred, or occluded CCTV faces
Traditional model	PCA + KNN, maximum 51.55% accuracy	Closed-set image classification baseline only
CNN	Two Conv2D blocks, dense 128, seven-class softmax	Identity classes are fixed; every new student requires classifier retraining
CNN result	100% training; approximately 82-86% validation	Strong overfitting signal; limited evaluation population
Unknown test	85% softmax threshold; 400 Olivetti faces all rejected	Cross-dataset appearance difference may cause easy rejection; not a calibrated open-set protocol
Software	Desktop application, database, dashboard, alerts	Consumes effort but does not create the central research contribution

1.3 What can be reused

- The broad surveillance motivation: low resolution, blur, pose, illumination, occlusion, compression, crowds, limited hardware, privacy, and real-time constraints.
- The high-level sequence of detection, alignment, feature extraction, matching, logging, and evaluation.
- The current implementation as a historical baseline for comparison, not as the final method.
- Existing report sections on standards, constraints, privacy, and institutional use, after rewriting them for the revised protocol.

1.4 What must be replaced

- The seven-class softmax recognition model.
- Haar Cascade as the final detector.
- LFW as the primary project dataset.
- Random image-level evaluation that does not separate identities or sessions.
- Softmax probability as an unknown-person detector.
- Desktop and mobile application work as core objectives.
- Claims of fairness without a defined demographic protocol and adequate sample sizes.

2. Final Singular Research Direction

Final title

DynaCampus-FR: Domain-Adaptive, Quality-Aware Open-Set Face Recognition with Training-Free Dynamic Enrollment for Low-Quality Campus CCTV

This is the only primary path. The project is not an attendance application, a mobile application, or a generic model-comparison exercise. It is a research study of dynamic open-set identity enrollment and recognition across the clean-photo-to-CCTV domain gap.

2.1 Central research problem

Universities have a changing population. A closed-set classifier assigns one output class to every enrolled person, so it becomes obsolete when students and faculty join or leave. At the same time, enrollment media are usually clean and frontal, while CCTV probes are distant, blurred, compressed, poorly illuminated, partially occluded, and temporally inconsistent. The project asks how to recognize enrolled identities and reject unknown identities under these conditions without retraining the recognition backbone every semester.

2.2 Proposed solution in one flow

```
Enrollment media
    -> SCRFD detection
    -> five-point alignment
    -> AdaFace embedding
    -> CCTV-aware enrollment variants
    -> quality-weighted identity prototype
    -> dynamic gallery

CCTV video
    -> frame sampling
    -> SCRFD face detection
    -> ByteTrack face tracks
    -> alignment
    -> AdaFace embedding
    -> residual CCTV Domain Adapter
    -> quality-aware multi-frame aggregation
    -> exact cosine gallery search
    -> KNOWN / UNKNOWN / AMBIGUOUS / INSUFFICIENT_QUALITY
```

2.3 Fixed decisions

Decision	Fixed choice	Reason
Primary recognition backbone	AdaFace IR-50 pretrained model	Quality-adaptive face recognition and manageable integration; primary research backbone [R3]
Detection	SCRFD	Efficient high-accuracy face detector for constrained compute [R1]
Baseline recognizer	ArcFace/InsightFace	Widely used strong embedding baseline [R2]
Tracking	ByteTrack applied to face detections	Simple, reproducible track construction using detection boxes [R4]
Main learnable component	Residual 512 -> 256 -> 512 domain adapter	Feasible within four months; backbone remains frozen
Enrollment	Dynamic gallery templates; no class retraining	Directly solves semester-wise enrollment
Primary gallery search	Exact cosine matrix search	Transparent and deterministic for the research population; no approximate-search confound
Scalable gallery extension	FAISS, optional	May be benchmarked after exact search is complete [R13]

Decision	Fixed choice	Reason
Primary local dataset	CampusFace-BD	Captures Bangladeshi campus conditions and dynamic enrollment
Primary public dataset	QMUL-SurvFace	Native low-resolution surveillance and open-set evaluation [R6]
Secondary transformer	FaceLiVTv2-S	Exploratory lightweight hybrid comparison; not the core method because it is a 2026 preprint [R10]
RAG	Excluded from core recognition	RAG does not improve biometric embeddings or identity decisions
Application	Minimal Streamlit research demo only	The paper and experiments, not software product features, are the priority

3. Research Motivation, Objectives, and Questions

3.1 Revised motivation

Educational institutions increasingly use CCTV, but automatic identification remains unreliable when faces are small, blurred, compressed, poorly illuminated, partially occluded, or non-frontal. More importantly, most student prototypes assume a fixed identity set and therefore require full retraining whenever the population changes. This is incompatible with semester-wise admission, graduation, faculty recruitment, and temporary campus visitors.

A practical research system should use an identity-independent embedding model, enroll a new person by inserting a template, process several frames from a track, distinguish unknown identities from low-quality evidence, and calibrate its operating threshold using open-set biometric metrics. The local research challenge is the clean enrollment versus CCTV probe domain shift in a Bangladeshi campus setting.

3.2 General objective

General objective
To develop and experimentally evaluate a domain-adaptive, quality-aware, open-set CCTV face-identification framework that supports training-free enrollment of new students and faculty under realistic low-resolution campus surveillance conditions.

3.3 Specific objectives

7. Create a consent-based, multi-session and multi-camera campus dataset with clean enrollment media, CCTV probes, known identities, unknown identities, and condition metadata.
8. Implement a modern pretrained baseline consisting of SCRFD detection, alignment, ArcFace/AdaFace embeddings, cosine matching, and dynamic gallery enrollment.
9. Develop CCTV-specific augmentation that reproduces low resolution, motion blur, defocus blur, noise, low illumination, compression, pose, and occlusion individually and in combination.
10. Train a lightweight residual Domain Adapter on development identities while freezing the pretrained recognition backbone.
11. Develop one-shot and few-shot CCTV-aware enrollment templates that can be inserted without model retraining.
12. Develop quality-aware temporal aggregation over tracked face sequences.
13. Calibrate a four-state open-set decision mechanism using score, top-1/top-2 margin, and quality.
14. Evaluate accuracy, open-set errors, robustness, demographic performance where valid, latency, memory, and model size.
15. Compare the primary CNN-based AdaFace method with a lightweight hybrid CNN-Transformer model after the core method is complete.

3.4 Research questions

ID	Research question
RQ1	How much do low resolution, motion blur, low illumination, compression, pose, and occlusion - individually and jointly - degrade modern campus CCTV face recognition?
RQ2	Can a lightweight residual CCTV Domain Adapter reduce the distance between clean enrollment embeddings and CCTV probe embeddings for unseen test identities?
RQ3	Does CCTV-aware augmented enrollment improve one-shot dynamic enrollment without retraining the recognition backbone?
RQ4	Does quality-aware multi-frame aggregation outperform single-frame, best-frame, and unweighted averaging strategies?
RQ5	Can score, gallery margin, and quality calibration improve open-set rejection compared with a similarity threshold alone?
RQ6	How does AdaFace compare with ArcFace and FaceLiVTv2-S under identical CCTV conditions and compute constraints?

3.5 Testable hypotheses

Hypothesis	Expected comparison
H1	AdaFace + Domain Adapter will produce higher TPIR at fixed FPIR than AdaFace without the adapter on identity-disjoint CCTV probes.
H2	A one-image CCTV-augmented enrollment template will outperform a single clean enrollment embedding.
H3	Quality-weighted track aggregation will outperform single-frame and unweighted mean aggregation.
H4	The four-state decision will reduce false-known assignments by separating unknown, ambiguous, and insufficient-quality cases.
H5	The hybrid transformer comparison may improve some degradation conditions, but its accuracy-efficiency trade-off must be measured rather than assumed.

4. Scope, Non-Goals, and Definition of Done

4.1 In scope

- Face detection, alignment, embedding extraction, track construction, dynamic enrollment, gallery matching, open-set rejection, and performance evaluation.
- A local consent-based dataset and a public surveillance benchmark.
- A small learnable adapter and reproducible ablation experiments.
- A minimal interface to demonstrate enrollment and video inference.
- Privacy, consent, withdrawal, access control, retention, and anonymization documentation.

4.2 Explicit non-goals

- Production campus deployment or replacement of human security staff.
- Automatic disciplinary action, attendance grading, or access denial based only on the model.
- A full desktop application, mobile application, or enterprise dashboard.
- Training a large face-recognition backbone from scratch.
- Deepfake detection, liveness detection, behavior analysis, emotion recognition, or weapon detection.
- RAG or an LLM as part of the biometric identity decision.
- Public release of identifiable face data without explicit participant and institutional authorization.

4.3 Project-level definition of done

16. A new identity can be enrolled by adding a gallery template without changing model weights.
17. The dataset has written consent, complete metadata, two CCTV sessions, genuine unknown identities, and frozen identity-disjoint splits.
18. The primary AdaFace pipeline, adapter, augmentation, aggregation, and four-state decision are reproducible from scripts and configuration files.
19. All principal ablations are run across three identity splits with fixed validation-only threshold selection.
20. Open-set metrics, robustness results, efficiency measurements, confidence intervals, and failure examples are reported.
21. The final report clearly distinguishes proven results, hypotheses, limitations, ethical boundaries, and future work.

5. Core Concepts and Terminology

Term	Meaning in this project
Face detection	Locating each face and producing a bounding box, confidence, and facial landmarks.
Face alignment	Geometrically normalizing a face using five landmarks so the eyes, nose, and mouth occupy consistent positions.
Embedding	A normalized numerical vector, usually 512 dimensions, representing identity-related facial features.
Gallery	The stored identity templates for enrolled students and faculty.
Probe	A face or track extracted from CCTV that must be searched against the gallery.
Verification	A 1:1 comparison asking whether two samples belong to the same person.
Identification	A 1:N search asking which gallery identity best matches a probe.
Closed set	Every probe identity is assumed to exist in the gallery.
Open set	Some probes are not enrolled and must be rejected as unknown.
Template	One normalized identity representation created from one or more embeddings.
Track	A sequence of face detections believed to belong to one moving person across frames.
Domain shift	The mismatch between clean enrollment media and degraded CCTV probes.
Dynamic enrollment	Adding or removing identity templates without retraining the recognition backbone.
FPIR	Fraction of non-mated searches that incorrectly produce one or more gallery candidates above threshold. [R15]

Term	Meaning in this project
FNIR	Fraction of mated searches that fail to return the correct identity above threshold. [R15]

$$cosine(a, b) = (a \cdot b) / (||a|| \; ||b||)$$

All embeddings are L2-normalized. For normalized vectors, cosine similarity equals their dot product.

PART II

METHOD AND ARCHITECTURE

Exactly how enrollment, CCTV processing, adaptation, aggregation, and open-set decisions work

6. End-to-End Architecture

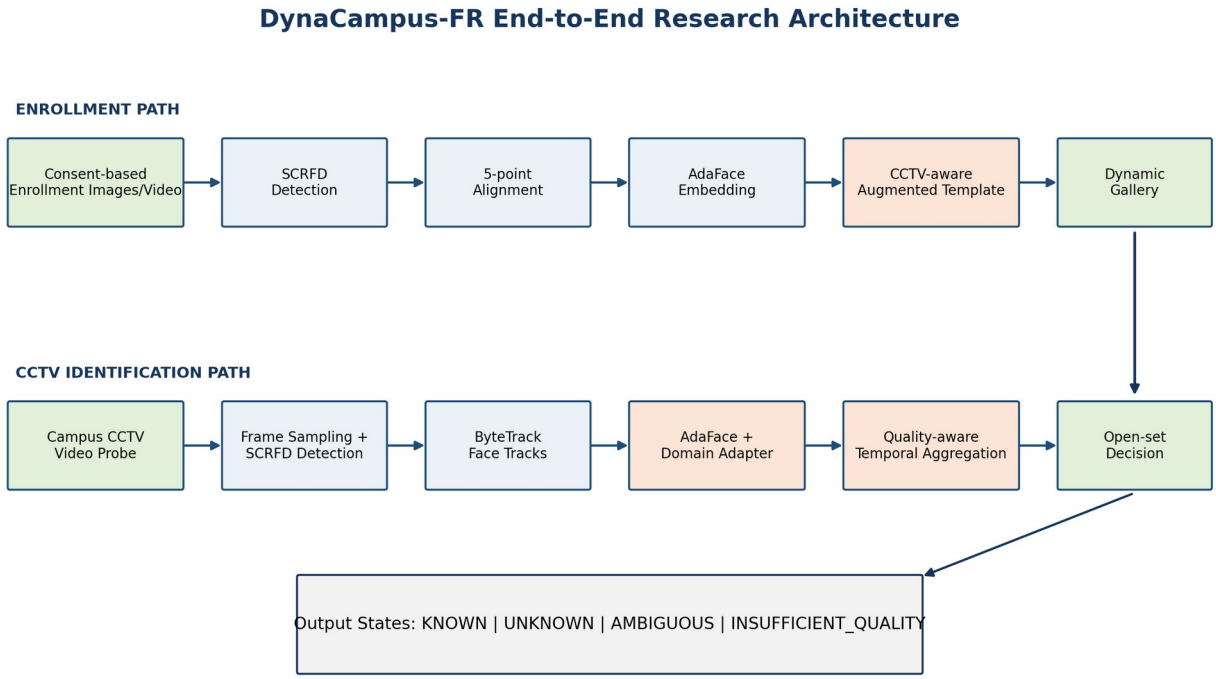


Figure 2. Full system architecture with separate enrollment and CCTV paths.

6.1 Enrollment path

22. Receive participant consent and an approved anonymous subject identifier.
23. Capture controlled enrollment photographs and a short enrollment video.
24. Detect faces with SCRFD and reject images with zero or multiple faces unless manually resolved.
25. Align each detected face to the recognizer input format, normally 112 x 112 pixels.
26. Extract normalized AdaFace embeddings.
27. Generate mild, identity-preserving CCTV-style enrollment variants.
28. Estimate quality and combine embeddings into an identity prototype.
29. Insert the prototype and subject ID into the gallery. No recognition backbone retraining occurs.

6.2 CCTV inference path

30. Read a recorded CCTV-style video or permitted live stream.
31. Sample frames at a configured rate; the initial dataset-processing rate is 5 FPS.
32. Detect all faces with SCRFD and retain bounding boxes, landmarks, confidence, and native face size.
33. Associate face detections over time using ByteTrack.
34. Align each usable crop and extract an AdaFace embedding.
35. Apply the residual CCTV Domain Adapter to the probe embedding.
36. Estimate frame quality and aggregate multiple embeddings per completed track.
37. Search the track template against all enrolled prototypes using exact cosine similarity.
38. Use calibrated score, top-1/top-2 margin, and quality thresholds to return one of four states.

7. Face Detection and Alignment

7.1 Detector choice: SCRFD

SCRFD is the fixed detector because it is designed for accuracy under constrained compute and is available through the InsightFace ecosystem. The project will begin with a pretrained SCRFD model and will not train a detector from scratch. Detector evaluation will still be performed on a manually checked local subset. [R1]

7.2 Detector inputs and outputs

```
Input:
    BGR or RGB video frame

Output per detected face:
    bbox = [x1, y1, x2, y2]
    detection_confidence
    five landmarks = left_eye, right_eye, nose, left_mouth, right_mouth
    native_face_width = x2 - x1
    native_face_height = y2 - y1
```

7.3 Detection operating rules

- Store detections before resizing so native face size remains available for quality analysis.
- Do not lower the detector threshold only to increase recall without measuring false detections.
- Use one detector configuration for all compared recognition models.
- Report detection recall by face-size category, not only overall recall.
- A detection with severe landmark failure may be retained for detector analysis but excluded from recognition.

7.4 Alignment

Apply a similarity transform using the five SCRFD landmarks and the target landmark template expected by the recognition model. The result is a normalized face crop. Alignment code must be shared across ArcFace, AdaFace, and FaceLiVTv2 unless the official model requires a documented alternative.

Important
Do not apply Canny, Sobel, or other edge extraction before the pretrained recognition model. Use aligned RGB/BGR images normalized exactly as required by the model repository.

8. Recognition Backbones

8.1 ArcFace baseline

ArcFace is the strong baseline embedding model. Its additive angular margin objective is widely used to produce discriminative face embeddings. The first two-week baseline may use the InsightFace buffalo_l package because it combines detection and recognition conveniently. [R2]

8.2 AdaFace primary backbone

AdaFace is the primary backbone. It adapts its training margin according to a sample-quality proxy and was designed to improve recognition under quality variation. The project uses a pretrained IR-50 model and freezes its weights. [R3]

8.3 FaceLiVTv2-S secondary comparison

FaceLiVTv2 is a 2026 lightweight hybrid CNN-Transformer preprint with an official repository. It is included only as a secondary comparison after the complete primary pipeline is stable. It must be described as a preprint rather than a peer-reviewed final standard. [R10]

Model	Role	Training in this project	Required output
ArcFace / InsightFace	Baseline	No backbone training	Normalized embedding
AdaFace IR-50	Primary method	No backbone training; adapter only	Normalized 512-D embedding
FaceLiVTv2-S	Secondary exploratory comparison	No backbone training unless official inference requires conversion	Normalized embedding mapped to the common evaluation interface

9. Training-Free Dynamic Enrollment

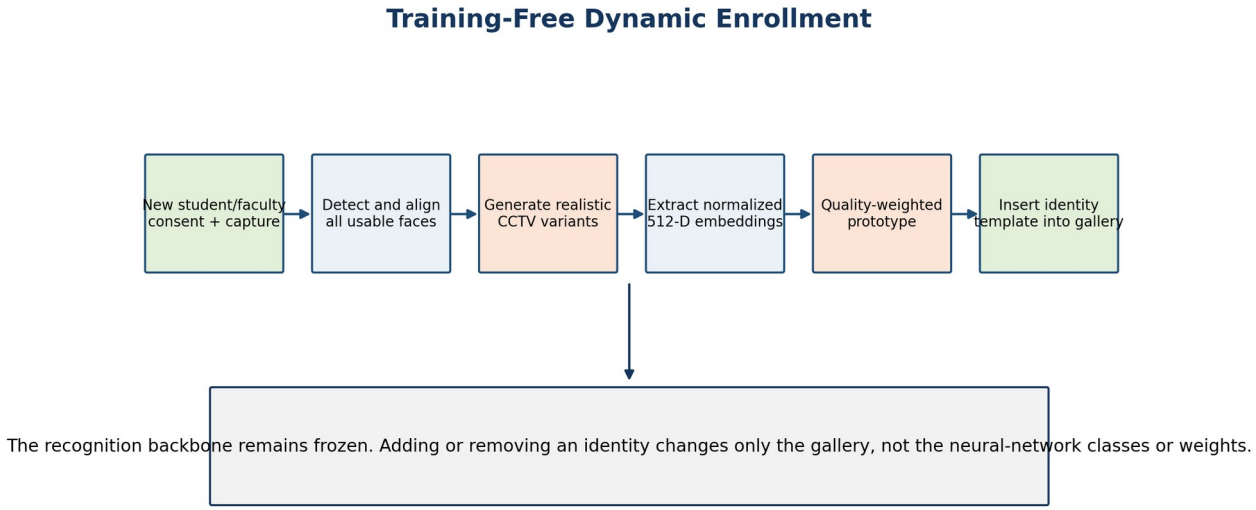


Figure 3. New identities are inserted as templates; the recognizer is not retrained.

9.1 Why the original classifier is infeasible

A softmax classifier has one output neuron per identity. When a new student arrives, the output layer and training dataset change. Retraining can be slow, requires earlier data, risks forgetting, and is operationally unnecessary because pretrained metric-learning models already support identity-independent embeddings.

9.2 Identity prototype

$$p_i = normalize(sum_k(q_k z_k) / sum_k(q_k))$$

z_k is a normalized embedding, q_k is its non-negative quality weight, and p_i is the identity template for subject i.

9.3 Enrollment modes to evaluate

Mode	Input	Purpose
E1: One clean image	One approved frontal photograph	Minimum-data baseline
E2: Five clean images	Frontal and mild pose images	Standard few-shot baseline
E3: One clean + CCTV augmentations	One clean image plus controlled synthetic degradations	Tests the proposed one-shot domain-aware enrollment
E4: Enrollment video	Top-quality frames from a short controlled video	Upper practical enrollment baseline
E5: Quality-weighted augmented template	Original, video, and mild CCTV variants with quality weighting	Proposed final enrollment method

9.4 Gallery operations

- INSERT: create a prototype and add it with a new anonymous subject ID.
- UPDATE: replace the prototype only after verified enrollment media or human-confirmed high-confidence observations.
- DEACTIVATE: exclude an identity from search while retaining an audit record.
- DELETE: remove the identity template and associated research records according to withdrawal or retention policy.

- VERSION: store template creation date, model version, adapter version, source media count, and quality summary.

10. CCTV-Aware Data Augmentation

10.1 Objective

Augmentation is used for two related purposes: training the small adapter to handle cross-domain degradation, and creating enrollment variants that reduce the mismatch between clean identity photographs and CCTV probes. It is not used to inflate test data or modify final evaluation images without disclosure.

10.2 Augmentation families and initial ranges

Family	Transform	Initial range or rule
Geometry	Horizontal flip	Probability 0.5 where identity semantics are preserved
Geometry	Rotation	Approximately -8 to +8 degrees
Geometry	Crop/translation	Small shifts that keep the full face visible
Illumination	Brightness/contrast	Mild to moderate changes; avoid full saturation
Illumination	Gamma	Initial gamma range approximately 0.6 to 1.4
Resolution	Downscale + upscale	Simulate native face widths around 16, 24, 32, 48, and 64 pixels
Blur	Motion blur	Kernel and direction varied; severity tiers
Blur	Gaussian/defocus blur	Mild, moderate, severe tiers
Compression	JPEG/image compression	Quality tiers approximately 15-80
Noise	Gaussian/sensor-like noise	Low, medium, high variance
Occlusion	Coarse dropout	Small regions; avoid removing the entire identity signal
Combined	Multiple degradations	Examples: low light + noise; downscale + compression; blur + low light + compression

10.3 Severity protocol

Level	Meaning	Use
0	Clean	Reference condition
1	Mild degradation	Likely campus CCTV but still recognizable
2	Moderate degradation	Challenging operational condition
3	Severe degradation	Stress test; may be below reliable recognition quality

10.4 Identity-preservation check

39. Visually inspect random augmented enrollment grids for every transform family.
40. Measure similarity between original and augmented embeddings; flag extreme outliers.

41. Do not use generative restoration or face synthesis in the core path because identity details may be altered.
42. Document every augmentation probability and parameter in YAML so results can be reproduced.

Albumentations is the fixed augmentation library because it supports composition of blur, compression, downscale, noise, and geometric transforms. [R11]

11. Residual CCTV Domain Adapter

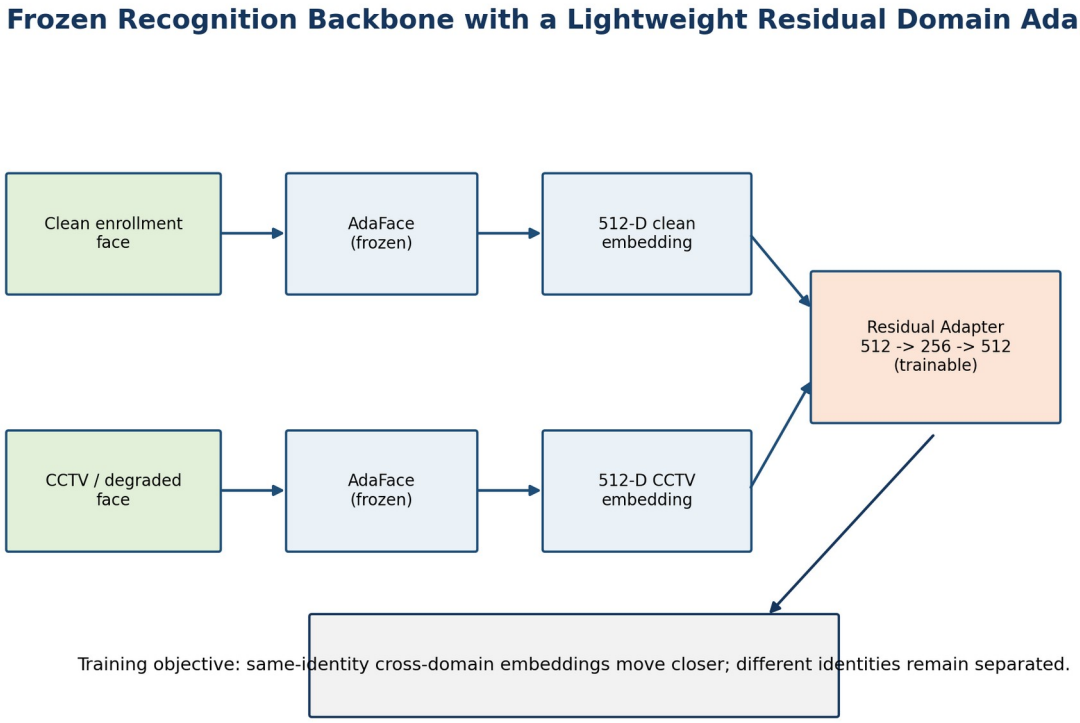


Figure 4. The recognition backbone is frozen; only a small residual adapter is trained.

11.1 Purpose

The adapter learns a small transformation of pretrained embeddings so that clean enrollment samples and CCTV probes from the same identity become closer, while different identities remain separated. Test identities never participate in adapter training.

11.2 Architecture

```
Input embedding: 512 dimensions
Layer 1: Linear(512, 256)
Activation: ReLU or GELU
Regularization: Dropout(0.1)
Layer 2: Linear(256, 512)
Residual output: z + adapter(z)
Final step: L2 normalization
```

$$z' = normalize(z + g_theta(z))$$

z is the frozen backbone embedding; g_theta is the trainable residual adapter.

11.3 Training samples

Triplet element	Source
Anchor	Clean enrollment embedding from a development identity
Positive	Real CCTV embedding or synthetic degraded embedding from the same development identity

Triplet element	Source
Negative	Embedding from a different development identity

11.4 Loss and initial hyperparameters

$$L_{triplet} = max(0, d(a,p) - d(a,n) + m)$$

The initial margin m is 0.2; cosine or Euclidean distance on normalized embeddings must be used consistently.

Setting	Initial value	Selection rule
Hidden dimension	256	Ablate 128 and 256 only if time permits
Optimizer	AdamW	Fixed unless instability is observed
Learning rate	1e-4	Select on development validation only
Batch size	64	Reduce if GPU memory is insufficient
Epochs	Maximum 30	Use early stopping
Early-stopping patience	5 epochs	Monitor validation loss and open-set development metric
Random seeds	42, 123, 2026	Report mean and variation if adapter training is stochastic

11.5 Critical leakage rule

No test identity leakage
The adapter may learn general CCTV transformation from development identities, but neither clean nor CCTV media from known-test or unknown-test identities can be used for adapter training, threshold selection, or hyperparameter tuning.

12. Tracking and Quality-Aware Temporal Aggregation

12.1 Why video must be treated as video

A CCTV person appears across many frames. Some frames are sharp, frontal, and large; others are blurred, occluded, or tiny. A single-frame decision ignores temporal redundancy and makes recognition unstable. ByteTrack will associate detections into track IDs so the system can form one representation per track. [R4]

12.2 Frame quality features

Feature	Interpretation	Initial computation
Detection confidence	Detector certainty	SCRFD output confidence
Face size	Available spatial detail	Normalized native face width/height
Sharpness	Motion/defocus blur	Normalized variance of Laplacian or alternative sharpness metric
Pose usability	Frontalness and landmark geometry	Approximate yaw/pitch from landmarks or landmark symmetry
Recognizer quality proxy	Model-derived evidence quality	Embedding norm or documented AdaFace proxy when available

12.3 Quality score policy

The quality score is a development-set calibration, not a universal truth. Start with normalized features and equal or simple weighted averaging, then select weights using development identities only. Store all component values so the final paper can explain why a frame was retained or rejected.

$$q_t = \sum_j (w_j f_{tj}), \text{ with } w_j >= 0 \text{ and } \sum_j (w_j) = 1$$

f_{tj} are normalized quality features for frame t. Weights are selected using the development set.

12.4 Aggregation methods

Method	Definition	Role
Single frame	First or predefined frame only	Weak baseline
Best frame	Embedding from highest-quality frame	Strong simple baseline
Mean	Unweighted average of all usable embeddings	Temporal baseline
Top-K mean	Mean of K highest-quality frames	Robust baseline
Quality-weighted mean	Weighted normalized average of embeddings	Proposed primary aggregation

$$z_{track} = normalize\left(\sum_t (q_t z_t) / \sum_t (q_t) \right)$$

The track representation is matched once against the gallery.

13. Open-Set Matching and Four-State Decision

13.1 Exact gallery search

For the core research population, stack all gallery prototypes in a matrix and compute the dot product with the normalized probe track embedding. This exact search is deterministic, transparent, and fast enough for tens or hundreds of identities. FAISS may later be used only for scalability experiments. [R13]

13.2 Decision variables

Variable	Meaning
s1	Highest cosine similarity between the probe track and any gallery template
s2	Second-highest similarity
m = s1 - s2	Gallery ambiguity margin
q_track	Aggregate track quality
tau_s	Minimum similarity threshold
tau_m	Minimum top-1/top-2 margin
tau_q	Minimum quality threshold

13.3 Four-state rule

```
if q_track < tau_q:
    return INSUFFICIENT_QUALITY
elif s1 < tau_s:
    return UNKNOWN
elif (s1 - s2) < tau_m:
    return AMBIGUOUS
else:
    return KNOWN(top1_identity)
```

13.4 Why insufficient quality is separate from unknown

A blurred or tiny face may belong to an enrolled person but contain too little evidence. Labeling it “unknown” would confuse identity absence with capture failure. The separate state supports honest evaluation and human review.

13.5 Threshold calibration

43. Generate mated and non-mated development searches.
44. Sweep candidate similarity, margin, and quality thresholds.
45. Select operating points that satisfy a target FPIR, initially 1% for the local dataset; report 0.1% only if there are enough unknown searches for a stable estimate.
46. Freeze thresholds before final test evaluation.
47. Report the entire trade-off curve, not only the selected operating point.

14. Vision Transformer and RAG Decisions

14.1 Vision Transformer placement

A transformer or hybrid model belongs in the face-embedding stage after detection and alignment. It does not replace the dataset, open-set protocol, dynamic gallery, or threshold calibration. FaceLiVTv2-S will use the same aligned face crops and evaluation splits as AdaFace so the comparison measures representation quality and efficiency rather than pipeline differences. [R10]

```
Frame -> SCRFD -> alignment -> FaceLiVTv2-S embedding
      -> same gallery/enrollment protocol
      -> same aggregation
      -> same open-set evaluation
```

14.2 RAG decision

RAG is excluded from the core project because retrieval-augmented generation is intended for retrieving external knowledge and generating text, not for producing or calibrating biometric identity embeddings. A later incident-summary assistant could retrieve anonymized event logs, but it must remain separate from identity decisions and is not part of the four-month core deliverables.

PART III

DATASET, ANNOTATION, AND ETHICS

How data will be obtained, protected, labeled, split, and evaluated without leakage

15. Dataset Strategy

The project requires more than one dataset role. Public datasets provide external comparability; the local dataset provides the clean-enrollment-to-campus-CCTV problem and dynamic enrollment scenario. The local dataset is the main contribution, while public data strengthens external validity.

Dataset	Role	Required or optional	Key facts
CampusFace-BD	Local dynamic enrollment and campus CCTV evaluation	Required	Consent-based; 60 target participants; multiple sessions/cameras; known and unknown identities
QMUL-SurvFace	Primary public low-resolution surveillance benchmark	Required if access is obtained	463,507 native surveillance face images of 15,573 identities; open-set focus [R6]
SCface	Small controlled surveillance-camera benchmark	Supporting	4,160 images, 130 subjects, five surveillance cameras [R8]
WIDER FACE	Detector context and optional detector evaluation	Supporting	32,203 images and 393,703 annotated faces with scale, pose, and occlusion variation [R7]
RFW	Demographic verification analysis	Supporting	Caucasian, Asian, Indian, and African testing subsets [R9]
IJB-C	Template/video protocol reference and possible stretch evaluation	Stretch	NIST challenge data include images, videos, and open/closed identification tasks [R14]

15.1 Public dataset use policy

- Obtain datasets only from official project pages or authorized request channels.
- Read and retain each dataset license or terms-of-use file.
- Do not mirror public datasets into the project repository.
- Use official protocols where practical; disclose any subset construction precisely.
- Do not use public benchmark test identities to train the local adapter.

16. CampusFace-BD Dataset Design

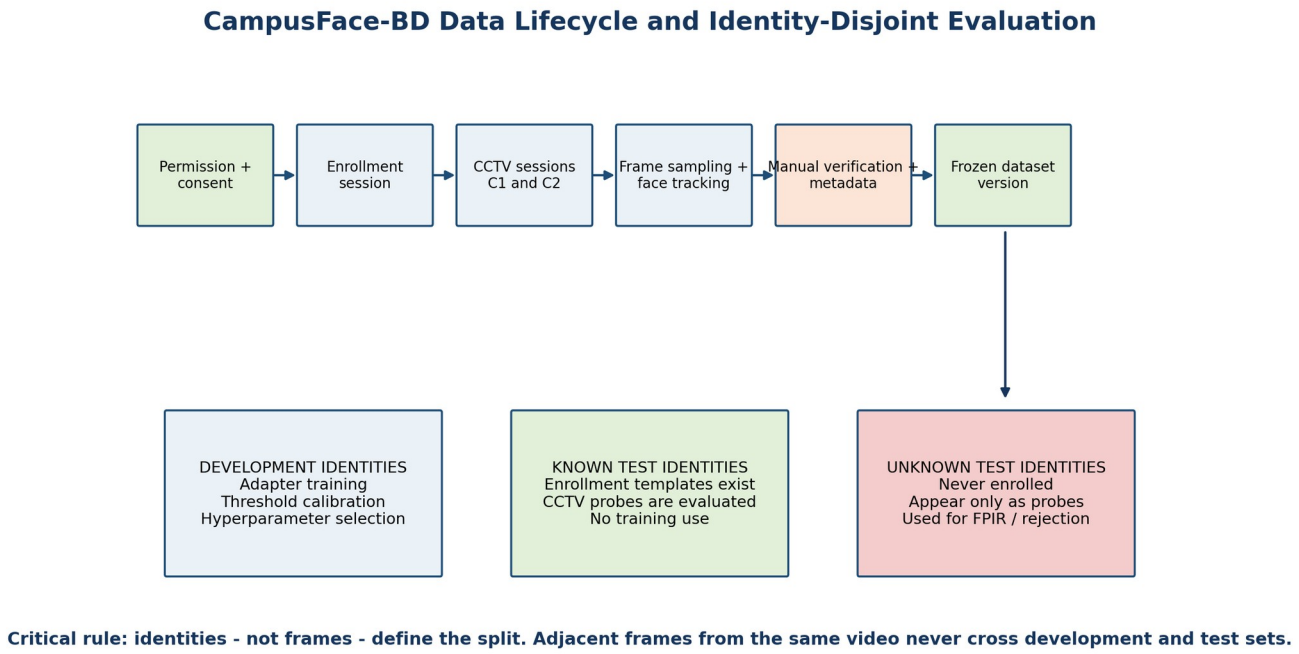


Figure 5. Local data lifecycle and identity-disjoint split.

16.1 Target scale

Item	Target
Participants	60 consenting adults associated with the institution
Enrollment photographs	15 per participant
Enrollment video	Approximately 15 seconds per participant
CCTV sessions	Two sessions on different days after enrollment
Cameras	Two fixed viewpoints
Primary distances	Near approximately 2 m; medium approximately 5 m; far approximately 8 m
Frame sampling	5 FPS for processed research frames
Manually checked detector frames	Approximately 1,000 representative frames
Expected processed face crops	Approximately 8,000-20,000 after quality control

16.2 Physical setup

- Use a controlled corridor or room where non-consenting bystanders can be excluded.
- Camera A: frontal or near-frontal corridor view.
- Camera B: diagonal or side view.
- Mount cameras approximately 2.3-2.8 meters high where feasible and record exact height, angle, resolution, frame rate, and codec.
- Use the same cameras across participants and document any camera change as a dataset version change.

16.3 Enrollment session

Capture	Count / duration
Frontal photographs	5
Slight left pose	3
Slight right pose	3
Slight upward pose	2
Slight downward pose	2
Slow head-movement video	Approximately 15 seconds

16.4 CCTV Session C1

- Normal indoor or daylight condition.
- Walk toward camera, away from camera, left-to-right, and right-to-left.
- Pause briefly at near, medium, and far positions.
- Record both cameras simultaneously where possible.

16.5 CCTV Session C2

- Record on a different day to create session variation.
- Use different clothing and naturally occurring appearance differences.
- Include lower illumination, mild glasses or mask condition when consented, partial occlusion, slightly faster walking, and a controlled small-group sequence.
- Do not force demographic or sensitive-attribute labeling that is not required by the research question.

17. Ethics, Consent, Privacy, and Governance

Mandatory precondition
No participant recording begins before supervisor approval, institutional permission, and a written consent process are complete.

17.1 Consent form content

- Research purpose and why face data are required.
- Types of enrollment and CCTV-style recordings.
- Who can access raw data and processed embeddings.
- How long data will be retained and when it will be deleted.
- Whether sample images may appear in reports or presentations.
- Whether the dataset will remain private or may be released later.
- How a participant can withdraw and what records will be deleted.
- Potential risks, including recognition errors and biometric sensitivity.

17.2 Identity management

Use anonymous IDs such as S001-S060. The real-name mapping is stored separately, encrypted, and accessible only to the authorized dataset lead and supervisor if required. Model outputs, tables, and plots use anonymous IDs.

17.3 Storage policy

Data type	Storage	Access
Signed consent forms	Encrypted private folder or locked institutional storage	Dataset lead and supervisor only

Data type	Storage	Access
Identity mapping	Encrypted file separated from images	Dataset lead only unless formally delegated
Raw videos and photos	Restricted Google Drive/institutional storage plus encrypted backup	Authorized research team
Processed face crops	Restricted research storage	Authorized research team
Embeddings	Restricted storage; treated as biometric templates	Authorized research team
Code and configs	Private GitHub during project; may later be public	Team and approved collaborators
Published examples	Only explicitly approved or strongly anonymized samples	Public

17.4 Operational boundaries

- No automated punishment, attendance penalty, or access denial.
- No claim that recognition is infallible or universally fair.
- Human review is required for ambiguous or consequential cases.
- Low-quality evidence must be rejected rather than forced into an identity.
- Template updates from CCTV require human verification to avoid gallery poisoning.

18. Naming, Folder Layout, and Metadata

18.1 Private dataset storage

```
CampusFaceBD_Data/
├── private/
│   ├── identity_map_encrypted.csv
│   └── consent_forms/
├── raw/
│   ├── enrollment/
│   ├── cctv_session_1/
│   └── cctv_session_2/
├── processed/
│   ├── frames/
│   ├── detections/
│   ├── aligned_faces/
│   ├── tracks/
│   └── embeddings/
├── annotations/
│   ├── cvat_exports/
│   └── verification/
├── metadata/
│   ├── participants.csv
│   ├── videos.csv
│   ├── frames.csv
│   ├── faces.csv
│   ├── tracks.csv
│   └── quality.csv
└── splits/
    ├── split_01.json
    ├── split_02.json
    └── split_03.json
```

18.2 Naming conventions

```
Enrollment photo:  S001_E_IMG_001.jpg
Enrollment video:  S001_E_VIDEO_001.mp4
CCTV Session 1:    S001_C1_CAMA_001.mp4
CCTV Session 2:    S001_C2_CAMB_001.mp4
Sampled frame:     F000001.jpg
Face crop:         FACE000001.jpg
```

```
Track:          T000001
```

18.3 Metadata schemas

File	Minimum fields
participants.csv	subject_id, consent_date, withdrawn, session_count, notes
videos.csv	video_id, subject_id, session_id, camera_id, resolution, fps, codec, lighting, occlusion, path
frames.csv	frame_id, video_id, timestamp, path
faces.csv	face_id, track_id, frame_id, bbox, landmarks, det_confidence, native_face_width, aligned_path
tracks.csv	track_id, subject_id, video_id, first_frame, last_frame, verified, known_status
quality.csv	face_id, size_score, blur_score, pose_score, model_score, final_quality

18.4 Dataset versioning

- Create a dataset version only after file names, metadata, verification, and splits are frozen.
- Record SHA-256 hashes for raw videos and split files where feasible.
- Never silently replace a file. Use a new version and changelog.
- Every result table must state the exact dataset version and split.

19. Annotation and Quality Control with CVAT

CVAT is the fixed annotation platform. It supports image and video annotation, tracking, quality control, and self-hosting. Track mode can create and edit annotations across sequences. [R12]

19.1 Annotation strategy

48. Run SCRFD automatically over all sampled frames.
49. Create face tracks automatically using ByteTrack.
50. Manually verify subject identity, false detections, missed faces, track fragmentation, and severe alignment errors.
51. Manually annotate approximately 1,000 representative frames in CVAT for detector evaluation.
52. Export annotations and preserve the original CVAT export with a version tag.

19.2 Annotation labels and attributes

Label / attribute	Values
face	Bounding box
subject_id	S001-S060 or UNKNOWN_CONTROLLED
occlusion	none / mild / moderate / severe
pose	frontal / moderate / extreme
distance	near / medium / far
illumination	normal / low / mixed
usable_for_recognition	yes / no / uncertain
track_verified	yes / no

19.3 Two-person verification rule

At least a sample of identity and track annotations should be independently checked by a second team member. Disagreements are resolved and recorded. The paper should report the number of frames/tracks checked and the correction rate.

20. Dataset Splits and Leakage Prevention

20.1 Three identity-disjoint splits

Group per split	Target identities	Use
Development	30	Adapter training, augmentation selection, threshold calibration, hyperparameter decisions
Known test	20	Have gallery enrollment media and independent CCTV probes; never used for training
Unknown test	10	Never enrolled; appear only as CCTV probes

20.2 Rotation example

Split 1 may use S001-S030 as development, S031-S050 as known test, and S051-S060 as unknown test. Splits 2 and 3 rotate identity groups so conclusions are not based on one favorable partition.

20.3 Session separation

- Enrollment media and CCTV probes are always separate capture domains.
- Where the protocol requires validation probes, use one CCTV session for development validation and the other for held-out analysis.
- Adjacent frames from the same video never appear in different identity groups or evaluation roles.
- Do not tune thresholds on the final known-test or unknown-test probes.

Most dangerous mistake
Randomly splitting individual frames from the same video into training and testing creates severe leakage because adjacent frames are nearly duplicates. Split by identity, session, and video, never by random frame.

PART IV

TECHNICAL IMPLEMENTATION

Platforms, software, repository, environment, scripts, and the exact order of implementation

21. Platforms and Tech Stack

Purpose	Fixed platform or technology
Code editing	VS Code
Version control	Git and private GitHub repository
Local environment	Windows 10/11 with Miniconda; optional WSL2 if required by a repository
GPU experiments	Google Colab GPU or an approved university GPU
Core language	Python 3.10
Deep learning	PyTorch, installed from the current official selector [R16]
Image/video	OpenCV
Detection baseline	InsightFace / SCRFD [R1]
Primary recognition	AdaFace official repository [R3]
Secondary recognition	FaceLiVT official repository [R10]
Augmentation	Albumentations [R11]
Tracking	ByteTrack [R4]
Exact gallery search	NumPy or PyTorch matrix multiplication
Optional scalable search	FAISS [R13]
Metrics and analysis	NumPy, Pandas, SciPy, scikit-learn, Matplotlib
Experiment logging	CSV, YAML, TensorBoard
Annotation	CVAT [R12]
Paper	Overleaf + Zotero
Architecture diagrams	diagrams.net
Minimal demo	Streamlit

22. Hardware and Storage

Resource	Minimum	Preferred
Local computer	8 GB RAM, 100 GB free storage	16 GB RAM or more, SSD
GPU	Not mandatory locally	Colab or NVIDIA GPU for adapter and comparison inference
Cameras	Two consistent recording devices	Two fixed cameras with known resolution/codec
Enrollment camera	Modern smartphone	Tripod and consistent indoor lighting
Backup	One restricted cloud copy	Restricted cloud plus encrypted external backup

22.1 Storage estimation

Reserve at least 100 GB before collection. Raw video size depends on resolution, codec, duration, and camera count. Processed frames and aligned faces can exceed raw-video size if stored without

compression. Keep raw media, derived faces, embeddings, and results in separate folders so unnecessary derivatives can be regenerated.

23. Repository Structure

```
DynaCampus-FR/
├── README.md
├── .gitignore
├── requirements.in
├── requirements-lock.txt
├── environment.yml
├── configs/
│   ├── paths.yaml
│   ├── models.yaml
│   ├── collection.yaml
│   ├── augmentation.yaml
│   ├── adapter.yaml
│   └── evaluation.yaml
├── notebooks/
│   ├── 00_environment_check.ipynb
│   ├── 01_face_detection_demo.ipynb
│   ├── 02_alignment_demo.ipynb
│   ├── 03_embedding_similarity_demo.ipynb
│   ├── 04_dynamic_enrollment_demo.ipynb
│   └── 05_open_set_demo.ipynb
├── src/
│   ├── detection/
│   ├── alignment/
│   ├── recognition/
│   ├── augmentation/
│   ├── tracking/
│   ├── quality/
│   ├── adapter/
│   ├── gallery/
│   ├── aggregation/
│   ├── open_set/
│   ├── evaluation/
│   └── utils/
├── scripts/
│   ├── extract_frames.py
│   ├── detect_faces.py
│   ├── build_tracks.py
│   ├── align_faces.py
│   ├── extract_embeddings.py
│   ├── prepare_adapter_data.py
│   ├── train_adapter.py
│   ├── build_gallery.py
│   ├── calibrate_thresholds.py
│   ├── evaluate_open_set.py
│   ├── evaluate_augmentations.py
│   ├── evaluate_aggregation.py
│   └── benchmark_runtime.py
├── tests/
├── results/
│   ├── raw/
│   ├── tables/
│   ├── figures/
│   ├── logs/
│   └── checkpoints/
└── docs/
    ├── research_protocol/
    ├── consent/
    ├── dataset_card/
    ├── experiment_log/
    └── paper_notes/
```

23.1 Repository rules

- No raw face images, videos, consent forms, or identity maps in Git.
- Configuration values belong in YAML, not scattered constants.
- Notebooks are for learning and exploration; final experiments run from scripts.
- Every experiment writes a configuration snapshot and result CSV.
- Large checkpoints remain in restricted storage unless redistribution is permitted.

```
# .gitignore core entries
data/
CampusFaceBD_Data/
```

```
*.mp4
*.avi
*.mov
*.jpg
*.jpeg
*.png
*.npy
*.npz
*.onnx
*.pt
*.pth
.env
__pycache__/
.ipynb_checkpoints/
```

24. Environment Setup from Zero

24.1 Accounts and software

53. Create or confirm GitHub, Google, Google Colab, Google Drive, Overleaf, and Zotero accounts.
54. Install Git, VS Code, Miniconda, and a current browser.
55. Install Docker Desktop only when the team is ready to self-host CVAT; CVAT Online may be used first if approved.

24.2 Repository and Conda environment

```
git clone <PRIVATE_REPOSITORY_URL>
cd DynaCampus-FR

conda create -n dynacampus python=3.10 -y
conda activate dynacampus
python -m pip install --upgrade pip
```

24.3 Core packages

```
pip install numpy pandas scipy scikit-learn matplotlib pillow pyyaml tqdm
pip install opencv-python albumentations tensorboard streamlit
pip install insightface onnxruntime

# For a CUDA Colab/runtime, use the compatible GPU runtime package:
pip install onnxruntime-gpu

# Install PyTorch using the command generated by the official selector:
# https://pytorch.org/get-started/locally/
```

Do not paste both CPU and GPU ONNX Runtime packages into the same final environment without checking compatibility. After the environment works, record exact versions:

```
pip freeze > requirements-lock.txt
conda env export --no-builds > environment.yml
```

24.4 Model repositories

```
# Clone official research repositories into a separate vendor/ or external/ folder.
git clone https://github.com/mk-minchul/adaface.git external/adaface
git clone https://github.com/FoundationVision/ByteTrack.git external/ByteTrack
# Clone FaceLiVT only in the later comparison phase.
git clone https://github.com/novendrastywn/FaceLiVT.git external/FaceLiVT
```

Licensing check
Before distributing code or model weights, record the license and usage restrictions of InsightFace pretrained models, AdaFace weights, FaceLiVT weights, ByteTrack code, and every dataset. Research use does not automatically permit commercial or unrestricted redistribution.

25. Configuration Files

25.1 models.yaml example

```
detector:
  name: scrfd
  input_size: [640, 640]
  confidence_threshold: 0.50

recognizer:
  primary: adaface_ir50
  baseline: insightface_arcface
  input_size: [112, 112]
  embedding_dim: 512

adapter:
  enabled: true
  hidden_dim: 256
  dropout: 0.10

tracker:
  name: bytetrack
  track_buffer: 30
```

25.2 augmentation.yaml example

```
severity_levels: [0, 1, 2, 3]
transforms:
  horizontal_flip:
    probability: 0.5
  rotation_degrees: 8
  downscale_face_widths: [16, 24, 32, 48, 64]
  jpeg_quality_ranges:
    mild: [60, 85]
    moderate: [30, 60]
    severe: [15, 35]
  motion_blur_kernel:
    mild: [3, 5]
    moderate: [5, 9]
    severe: [9, 15]
combined_profiles:
  - low_resolution_plus_compression
  - low_light_plus_noise
  - blur_plus_low_light_plus_compression
```

25.3 evaluation.yaml example

```
splits: [split_01, split_02, split_03]
seeds: [42, 123, 2026]
open_set_targets:
  fpir: [0.01, 0.001]
verification_targets:
  far: [0.01, 0.001]
aggregation_methods:
  - single_frame
  - best_quality
  - mean
  - top_k_mean
  - quality_weighted
report_runtime: true
bootstrap_samples: 1000
```

26. Implementation Order for a Complete Beginner

Rule

Do not begin with training. First make a pretrained detector and embedding pipeline work on a few known and unknown people.

26.1 Stage A - one-image detection

56. Place one clear image in a temporary local folder.
57. Run InsightFace FaceAnalysis or the SCRFD inference wrapper.
58. Draw bounding boxes and landmarks.
59. Save the output image and print the number of detected faces.
60. Repeat with a group image and a low-light image.

26.2 Stage B - embedding similarity

61. Prepare two images of Person A and one image of Person B.
62. Detect one face in each image, align it, and extract normalized embeddings.
63. Compute cosine similarity between A1-A2 and A1-B1.
64. Save scores to a CSV. Do not select a production threshold from three images.

26.3 Stage C - five-person gallery

65. Collect three consented development images from each team member for a private technical demo.
66. Create one mean template per person.
67. Search a new photo against the five templates.
68. Add a sixth person by inserting a new template. Confirm that no model training occurs.

26.4 Stage D - video and track aggregation

69. Record a short test video with one person moving.
70. Detect faces in frames and collect embeddings.
71. Compare first-frame, best-frame, and average embedding matches.
72. Only after this works should ByteTrack and the full quality model be integrated.

27. Core Interfaces and Pseudocode

27.1 Enrollment function

```
def enroll(subject_id, media_paths, recognizer, quality_model, gallery):
    samples = []

    for path in media_paths:
        image = load_image(path)
        face = detect_single_approved_face(image)
        aligned = align_face(image, face.landmarks)
        quality = quality_model(aligned, face)

        if quality < ENROLLMENT_MIN_QUALITY:
            continue

        embedding = l2_normalize(recognizer(aligned))
        samples.append((embedding, quality))

    augmented_samples = create_cctv_enrollment_variants(samples)
    prototype = quality_weighted_template(samples + augmented_samples)
    gallery.insert(subject_id, prototype, metadata=...)
    return prototype
```

27.2 Video identification

```
for frame in video_stream:
    detections = detector(frame)
    tracks = tracker.update(detections)

    for track in tracks:
        aligned = align_face(frame, track.landmarks)
```

```
        embedding = recognizer(aligned)
        adapted = adapter(embedding)
        quality = estimate_quality(aligned, track)
        track.add(adapted, quality)

for track in completed_tracks:
    probe = aggregate(track.embeddings, track.qualities)
    top1_id, s1, s2 = exact_gallery_search(probe)
    decision = open_set_rule(s1, s2, track.quality)
    save_result(track, decision, top1_id, s1, s2)
```

27.3 Exact gallery search

```
# gallery_matrix shape: [N_identities, embedding_dim]
# probe shape: [embedding_dim]
# all vectors are L2-normalized
scores = gallery_matrix @ probe
ranked = scores.argsort(descending=True)
top1_index = ranked[0]
top2_index = ranked[1]
s1 = scores[top1_index]
s2 = scores[top2_index]
```

28. Reproducibility and Coding Rules

- Set and log Python, NumPy, and PyTorch random seeds.
- Record Git commit hash, dataset version, split, model weight name, adapter checkpoint, and configuration file for every run.
- Use relative paths resolved from a single paths.yaml file.
- Use type hints, docstrings, and explicit error messages in final scripts.
- Write unit tests for cosine matching, template normalization, split validation, and threshold decision logic.
- Save raw predictions before computing metrics so plots can be regenerated without rerunning models.
- Never overwrite final results; create timestamped or experiment-ID folders.

```
experiment_id,date,git_commit,dataset_version,split,seed,backbone,adapter,
augmentation,enrollment,aggregation,threshold_version,notes

EXP001,2026-08-10,abc123,CAMPUS_V1,split_01,42,AdaFace,off,
none,one_shot,single_frame,T01,baseline
```

PART V

EXPERIMENTS AND EVALUATION

The exact comparisons, metrics, calibration rules, and result artifacts required for a publishable study

29. Core Experiment Matrix

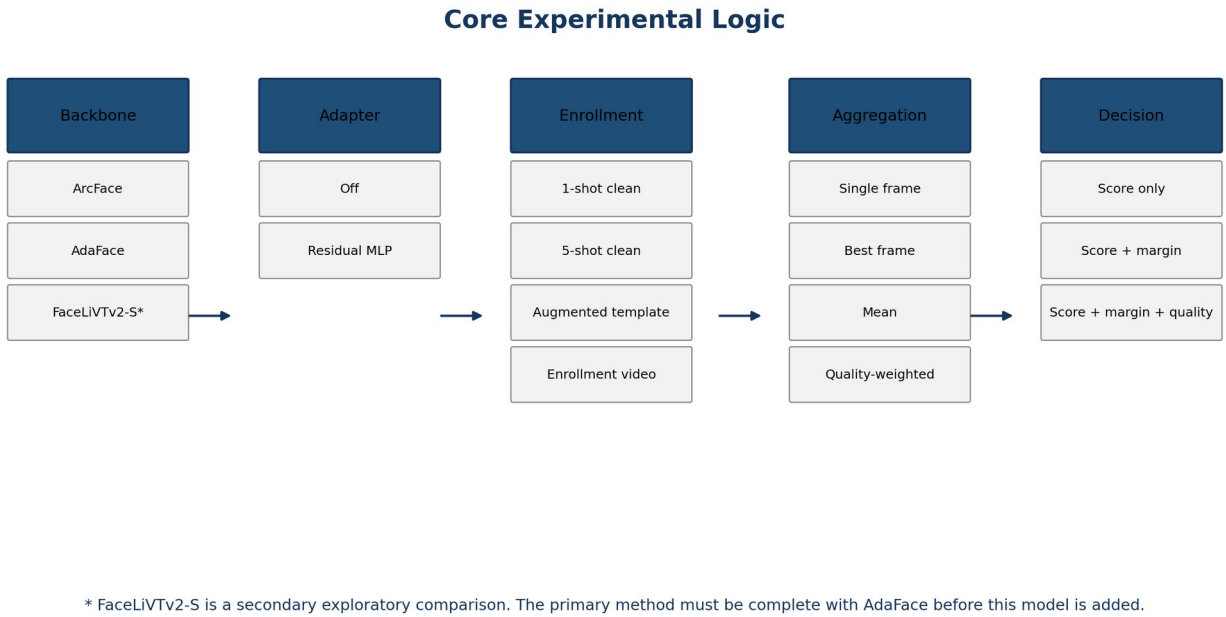


Figure 6. Main factors varied in the experimental program.

Experiment	Comparisons	Primary question
A. Backbone	ArcFace vs AdaFace vs FaceLiVTv2-S	Which representation gives the best CCTV accuracy-efficiency trade-off?
B. Domain Adapter	AdaFace without vs with residual adapter	Does cross-domain adaptation generalize to unseen identities?
C. Augmentation	None vs basic vs CCTV-specific vs combined	Which degradation training improves robustness?
D. Enrollment	One-shot clean vs five-shot vs augmented vs video vs proposed template	How much enrollment data is necessary?
E. Aggregation	Single vs best vs mean vs Top-K vs quality-weighted	Does temporal quality weighting improve identity search?
F. Decision	Score only vs score+margin vs score+margin+quality	Can ambiguity and low quality reduce false-known outputs?
G. Conditions	Distance, illumination, blur, pose, occlusion, resolution	Where does the system fail?
H. Runtime	Latency, FPS, memory, model size, insertion/search time	Is the method deployable on constrained resources?

30. Required Ablation Sequence

73. ArcFace exact-search baseline.
74. AdaFace exact-search baseline.
75. AdaFace + one-shot clean enrollment.
76. AdaFace + CCTV-augmented enrollment.
77. AdaFace + Domain Adapter.
78. AdaFace + Domain Adapter + quality-weighted aggregation.
79. Full four-state open-set decision.
80. FaceLiVTv2-S with the same final enrollment and decision protocol.

Each added component must be compared with the immediately preceding system. Otherwise, the paper cannot show which component produced improvement.

31. Metrics

31.1 Detection metrics

- Precision and recall on the manually annotated local subset.
- Average Precision where the annotation protocol supports it.
- Recall by native face-size group: small, medium, large.
- Miss and false-detection examples by condition.

31.2 Verification metrics

- ROC-AUC and Equal Error Rate.
- TAR at FAR = 1% and FAR = 0.1% when sample size permits.
- FNMR at specified FMR, consistent with biometric reporting conventions. [R15]

31.3 Open-set identification metrics

- Rank-1 and Rank-5 identification rate for mated probes.
- TPIR or 1-FNIR at FPIR = 1%.
- TPIR or 1-FNIR at FPIR = 0.1% only when the number of non-mated searches is adequate.
- Full FNIR versus FPIR curve.
- Unknown rejection rate and false-known rate.

NIST defines FNIR as the proportion of mated searches failing to return the correct identity above threshold and FPIR as the proportion of non-mated searches producing one or more candidates above threshold. [R15]

31.4 Quality-rejection metrics

- Coverage: fraction of tracks that receive a known/unknown decision rather than insufficient quality.
- Accuracy or FNIR as a function of coverage.
- Error rate before and after the quality gate.

31.5 Efficiency metrics

- Detection latency per frame.
- Recognition and adapter latency per aligned face.
- End-to-end effective FPS.
- Peak CPU RAM and GPU memory.
- Model parameter count and checkpoint size.
- Gallery insertion and exact search time as gallery size increases.

32. Threshold Calibration Procedure

81. Produce development mated searches from enrolled development identities and independent CCTV probes.
82. Produce development non-mated searches using identities that are not in the relevant gallery.
83. Save s1, s2, margin, quality, and ground-truth status for every track.
84. Sweep tau_s, tau_m, and tau_q on a grid.
85. Select thresholds based on a predeclared target FPIR and a secondary objective such as minimum FNIR.
86. Freeze threshold version T01, T02, and so on; do not silently change it after viewing final test results.
87. Apply the same frozen thresholds to all final test identities for that model/configuration.

No test-set tuning
Any threshold chosen after viewing final test labels invalidates the claimed held-out evaluation. If a threshold must change, the final test must be rerun under a newly declared protocol.

33. Statistical Analysis

- Report mean and standard deviation across three identity-disjoint splits.
- For adapter training, report multiple seeds if results vary materially.
- Use 1,000 bootstrap resamples of probe tracks or identities for 95% confidence intervals.
- Use paired bootstrap or another paired test when comparing two methods on the same probes.
- Do not claim a meaningful improvement from a very small numerical difference without uncertainty analysis.
- Report the number of mated and non-mated searches at every operating point.

34. Condition-Wise Analysis

Factor	Categories
Distance	Near / medium / far
Native face width	<24 px / 24-47 px / 48-95 px / >=96 px, adjusted after data inspection
Illumination	Normal / low / mixed
Pose	Frontal / moderate / extreme
Occlusion	None / mild / moderate / severe
Blur	Low / medium / high
Compression	Native codec groups or controlled synthetic levels
Session	C1 / C2
Camera	A / B

Condition labels may be assigned per video, track, or frame depending on the factor. The unit must be specified. Avoid pretending that an automatically estimated pose or blur category is perfect ground truth.

35. Fairness and Demographic Analysis

The source report is concerned about regional and skin-tone bias. The revised study must not make unsupported fairness claims. Use RFW for a standard public demographic verification check and, if local demographics are collected, do so only with ethical approval and sufficient sample sizes. [R9]

- Report subgroup error rates, not only overall accuracy.
- Do not infer ethnicity or sensitive attributes automatically from faces.
- If a subgroup has too few participants, state that reliable analysis is impossible.
- Separate demographic differences from capture-condition differences where possible.

36. Required Result Artifacts

Artifact	Minimum content
Table 1	Dataset statistics by session, camera, distance, and identity group
Table 2	Backbone comparison with accuracy and efficiency
Table 3	Adapter ablation
Table 4	Enrollment strategy comparison
Table 5	Aggregation comparison
Table 6	Open-set decision ablation

Artifact	Minimum content
Table 7	Condition-wise results
Figure 1	Complete architecture
Figure 2	Dataset collection and split protocol
Figure 3	ROC verification curve
Figure 4	FNIR-FPIR open-set curve
Figure 5	Performance versus native face size
Figure 6	Coverage versus error for quality rejection
Figure 7	Representative success and failure cases

37. Failure Analysis

88. Select true-known successes, false-unknowns, false-knowns, ambiguous cases, and insufficient-quality cases.
89. For each case, show or record face size, blur, pose, occlusion, camera, track length, s1, s2, margin, and quality.
90. Separate detection failure, tracking failure, alignment failure, embedding mismatch, and threshold failure.
91. Check whether augmented enrollment or the adapter helped or harmed the case.
92. Do not publish identifiable examples without explicit permission.

PART VI

FOUR-MONTH EXECUTION PLAN

The exact order, weekly deliverables, team responsibilities, gates, and contingency actions

38. Sixteen-Week Roadmap

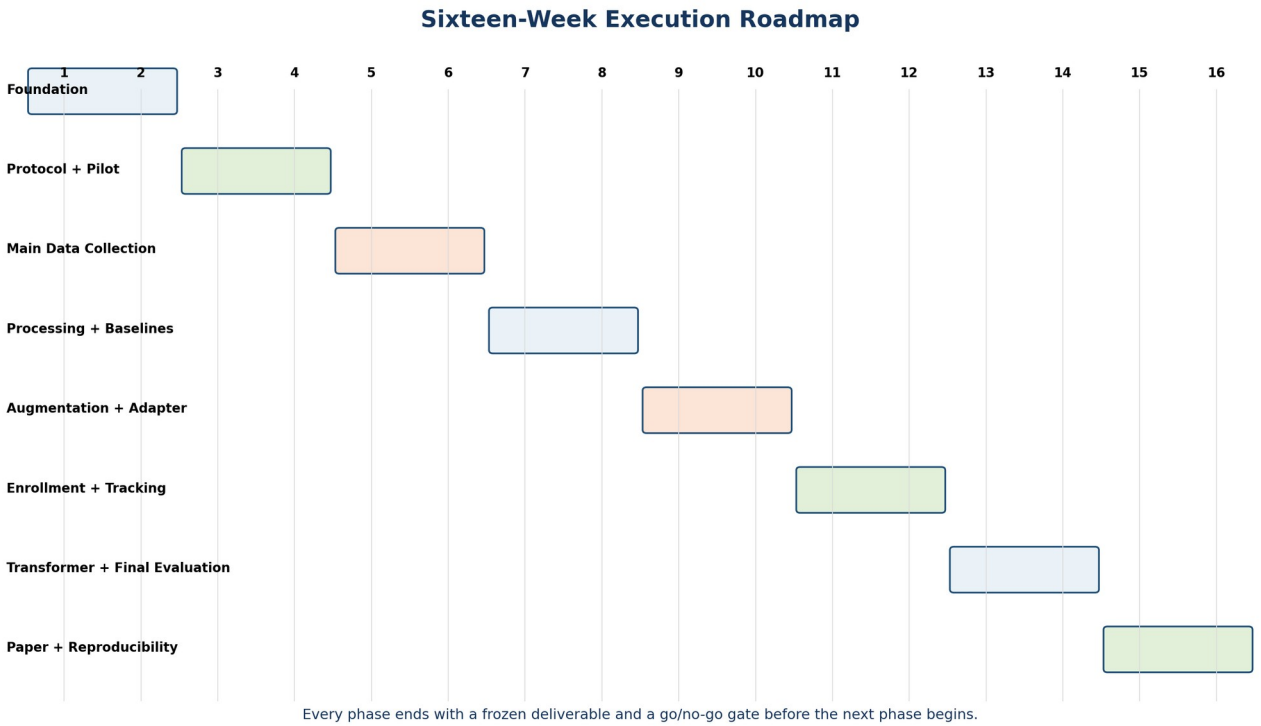


Figure 7. Four-month roadmap. Each phase has a frozen deliverable.

Week	Focus	Main work	Exit deliverable
1	Environment and face detection	Repository, Conda environment, SCRFD/InsightFace detection notebook	Face boxes and landmarks saved for clear, group, and low-light images
2	Embeddings and dynamic gallery	Cosine similarity, five-person gallery, sixth-person insertion without retraining	Working known/unknown technical baseline
3	Research protocol and ethics	Final title, questions, consent, permission, metadata templates	Written approval path and frozen collection protocol
4	Pilot dataset	Five participants, two sessions, two cameras, end-to-end pilot	Pilot report; protocol corrections
5	Main collection I	First 30 participants and enrollment/C1 data	Daily backup and metadata audit
6	Main collection II	Remaining participants and C2 recordings	Raw 60-participant dataset target
7	Processing and annotation	5 FPS frames, SCRFD, tracks, alignment, CVAT subset, verification	CampusFace-BD processed version 1
8	Public and local baselines	ArcFace and AdaFace baseline, public data setup, threshold-development files	Baseline tables and curves
9	CCTV augmentation	Severity profiles, identity-preservation checks, augmentation experiments	Augmentation ablation
10	Domain Adapter	Triplet preparation, training, early stopping, before/after comparison	Frozen best adapter checkpoint
11	Dynamic enrollment research	One-shot, five-shot, augmented, video, quality-weighted templates	Enrollment comparison table
12	Tracking and aggregation	ByteTrack, best-frame, mean, Top-K, quality-weighted	Temporal aggregation table

Week	Focus	Main work	Exit deliverable
		aggregation	
13	Transformer comparison	FaceLiVTv2-S inference under identical protocol	CNN versus hybrid efficiency table
14	Final evaluation	Three splits, confidence intervals, condition-wise and runtime results	Frozen final results; no casual tuning
15	Paper writing	Methods, data, experiments, results, discussion, ethics, limitations	Complete draft with all tables and figures
16	Reproducibility and demo	Clean rerun, lockfile, README, dataset card, Streamlit demo, presentation	Final report and handoff package

39. First Seven Days - Start Now

Day	Tasks	Definition of done
1	Install Git, VS Code, Miniconda; create private repository and folder structure	Initial commit pushed
2	Create Python environment; install packages; verify PyTorch, OpenCV, ONNX Runtime	00_environment_check.ipynb passes
3	Run SCRFD/InsightFace on one face, group image, and low-light image	01_face_detection_demo.ipynb saves boxes and landmarks
4	Extract embeddings for same and different persons; save similarity CSV	03_embedding_similarity_demo.ipynb works
5	Build a five-person gallery from three consented images each	Known test images retrieve correct top candidates
6	Add a sixth identity without training; measure insertion and search time	Dynamic enrollment demonstrated
7	Run a 20-second video; compare first-frame and average-embedding scores	First video-level recognition notebook complete

End-of-week statement
“The recognition backbone remains fixed. A new identity is enrolled by creating and inserting a template, not by retraining a class-based CNN.”

40. Team Task Allocation

Role	Primary responsibilities	Secondary responsibilities
Dataset and Ethics Lead	Permission, consent, scheduling, identity map, metadata, backups	Dataset card and withdrawal handling
Detection and Tracking Lead	SCRFD, frame extraction, alignment, ByteTrack, CVAT	Detector evaluation and track QA
Recognition Lead	ArcFace, AdaFace, FaceLiVTv2, embedding interface, gallery	Model licensing and runtime benchmark
Method Lead	Augmentation, adapter, quality, temporal aggregation	Ablations and failure analysis
Evaluation and Paper Lead	Splits, metrics, threshold calibration,	Reproducibility audit and presentation

Role	Primary responsibilities	Secondary responsibilities
	statistics, tables, Overleaf	

All members participate in data verification, code review, and paper review. No single person should be the only one who understands a critical module.

41. Milestone Gates

Gate	Question	If the answer is no
G1 - Baseline	Can the team detect, align, embed, compare, and dynamically enroll a new identity?	Do not collect the main dataset; fix the baseline
G2 - Ethics	Are permission, consent, storage, and withdrawal processes approved?	Do not record participants
G3 - Pilot	Does the pilot produce usable faces at all planned distances and both cameras?	Change setup before main collection
G4 - Dataset freeze	Are metadata, identities, tracks, and splits verified?	Do not train the adapter
G5 - Primary method	Are AdaFace baseline, adapter, enrollment, aggregation, and open-set decisions complete?	Do not add FaceLiVTv2
G6 - Final evaluation	Are thresholds frozen and all splits reproducible?	Do not write final claims
G7 - Submission	Can results be regenerated from a clean environment?	Fix dependencies and scripts

42. Risk Register and Contingency Plan

Risk	Likelihood / impact	Prevention	Contingency
Institutional permission delayed	Medium / High	Submit protocol in Week 3	Continue public benchmark and five-person technical pilot only; do not collect unauthorized data
Unable to recruit 60 participants	Medium / Medium	Schedule early and recruit in batches	Minimum acceptable target 40 with stronger public evaluation; report limitation
Far-distance faces too small	High / High	Pilot camera placement and resolution	Adjust distance categories or camera placement before main collection
Public dataset access delayed	Medium / Medium	Request in Week 1-2	Use SCface and local data while waiting; keep QMUL integration modular
AdaFace repository incompatibility	Medium / Medium	Freeze a working environment early	Use a verified AdaFace inference wrapper or ArcFace baseline temporarily; do not abandon embedding approach
ByteTrack fails on face-only detections	Medium / Medium	Test in pilot videos	Use a simple IoU + embedding association tracker as documented fallback
Adapter overfits	Medium / High	Identity-disjoint development/test, early stopping, small model	Reduce capacity; use stronger regularization; report negative result if necessary
Augmentation damages identity	Medium / High	Visual and embedding checks	Remove aggressive transforms;

Risk	Likelihood / impact	Prevention	Contingency
			keep only validated ranges
Threshold unstable at 0.1% FPIR	High / Medium	Collect many non-mated searches	Report 1% FPIR as primary and label lower rates exploratory
Team spends time on interface	High / Medium	Freeze app scope	Build Streamlit only after final results
Privacy breach	Low / Critical	Restricted storage, encryption, access list	Stop access, notify supervisor, follow institutional incident procedure

PART VII

PAPER, REPRODUCIBILITY, AND HANDOFF

How the final research story, deliverables, limitations, and AI context should be presented

43. Intended Research Contributions

93. CampusFace-BD: a consent-based multi-session, multi-camera campus CCTV dataset and dynamic open-set protocol.
94. A training-free enrollment framework that adds and removes identities through quality-controlled templates rather than classifier retraining.
95. A lightweight residual CCTV Domain Adapter that reduces clean-enrollment-to-CCTV embedding mismatch for unseen identities.
96. CCTV-aware augmented one-shot enrollment for limited institutional photographs.
97. Quality-aware multi-frame aggregation and a calibrated four-state open-set decision.
98. A controlled comparison of ArcFace, AdaFace, and a lightweight hybrid transformer under identical surveillance conditions and efficiency measurements.

43.1 Claim discipline

The final paper may claim only contributions supported by measured results. If the adapter does not improve performance, the paper should report the negative result and emphasize the dataset, protocol, dynamic enrollment, and temporal analysis rather than inventing an improvement.

44. Paper Structure

Section	Required content
Abstract	Problem, method, dataset, primary quantitative result, significance, and limitation
1. Introduction	Changing campus population, CCTV domain gap, open-set need, research questions, contributions
2. Related Work	Surveillance recognition, open-set identification, low-quality recognition, quality-aware aggregation, dynamic enrollment, hybrid models
3. Dataset and Protocol	Ethics, participants, cameras, sessions, conditions, metadata, splits, public datasets
4. Proposed Method	Detection, alignment, backbone, adapter, enrollment, quality, aggregation, four-state decision
5. Experimental Setup	Models, versions, hyperparameters, metrics, calibration, hardware, statistics
6. Results	Main table, ablations, open-set curves, condition analysis, runtime, transformer comparison
7. Discussion	Interpretation, failures, fairness, operational boundaries, negative results
8. Ethics and Limitations	Consent, privacy, sample size, dataset access, domain limits, no automated consequential action
9. Conclusion	What was learned and the next research steps

45. Minimal Research Demo

The final Streamlit demo is evidence of integration, not the research contribution. It should have only the following functions:

- Enroll a new anonymous identity from approved images or a short video.
- Show that the gallery updates without model retraining.
- Upload a permitted CCTV-style video and display tracks.
- Display KNOWN, UNKNOWN, AMBIGUOUS, or INSUFFICIENT_QUALITY.
- Show top similarity, second similarity, margin, track quality, and model/version metadata.

- Export an anonymized experiment log.

Do not build messaging, notifications, mobile UI, attendance grading, account roles, or complex analytics until the paper is complete.

46. Final Reproducibility Package

Deliverable	Contents
README.md	Setup, environment, folder structure, commands, datasets, model acquisition, reproduction steps
requirements-lock.txt / environment.yml	Exact dependencies
configs/	Every final YAML configuration
splits/	Frozen identity split files without real names
scripts/	Data processing, training, calibration, evaluation, runtime
results/raw/	Per-probe predictions, scores, quality, ground truth
results/tables/	Final CSV and LaTeX tables
results/figures/	Publication figures regenerated from raw results
dataset card	Purpose, composition, collection, consent, intended use, prohibited use, limitations
model card	Models, versions, training, thresholds, conditions, limitations
experiment log	Experiment IDs, Git hashes, configs, seeds, notes

47. Limitations That Must Be Stated

- The local dataset population and institution are limited; results may not generalize to all universities or demographics.
- Camera placement and recording protocol are controlled compared with unrestricted operational CCTV.
- The system uses pretrained models whose original training data and licensing impose constraints.
- The adapter is small and may not solve severe information loss in tiny faces.
- Open-set thresholds depend on gallery size, capture conditions, and target FPIR.
- Quality rejection improves safety by refusing some decisions but reduces coverage.
- Face recognition has privacy, fairness, and governance risks and must not be treated as an unquestionable authority.
- FaceLiVTv2 is a preprint comparison and may change after later review or repository updates.

48. Statements Future Assistants Must Not Contradict

Source-of-truth constraints
The project is research-first; the core recognizer is embedding-based; new identities are added without retraining; the primary model is AdaFace; the detector is SCRFD; the local dataset is consent-based; the evaluation is identity-disjoint and open-set; applications and RAG are not core contributions.

- Do not propose a seven-class or N-class softmax CNN as the final recognition system.
- Do not use LFW as the main campus surveillance dataset.
- Do not use softmax confidence as the only unknown-person rule.
- Do not random-split adjacent frames.
- Do not collect data without permission and consent.
- Do not optimize the app before the experiments.

- Do not add extra research tasks unless they support the six research questions and fit the frozen timeline.

49. Startup Prompt for Any AI/LLM Assistant

The following block can be copied together with this PDF when starting a new AI conversation:

```
You are assisting with DynaCampus-FR, a four-month research-first FYDP.
Treat the attached Complete Project Context as the source of truth.

The project studies domain-adaptive, quality-aware open-set face recognition
for low-quality campus CCTV with training-free dynamic enrollment.

Before suggesting code or changes:
1. Identify the current week, milestone, and deliverable.
2. Follow the fixed architecture and repository structure.
3. Preserve identity-disjoint data splits and validation-only threshold tuning.
4. Do not expose or request public upload of biometric data.
5. Do not replace the embedding gallery with a closed-set classifier.
6. Separate required core work from optional extensions.
7. Provide exact file paths, commands, expected inputs, expected outputs,
   validation checks, and failure handling.
8. State any assumption or dependency that is not present in the context file.

The core path is:
SCRFD -> alignment -> AdaFace -> residual CCTV Domain Adapter ->
quality-aware track aggregation -> exact cosine gallery ->
four-state open-set decision.

ArcFace is the baseline. FaceLiVTv2-S is a late secondary comparison.
RAG and full applications are outside the core scope.
```

50. Final Master Checklist

☐ Repository created and private
☐ Working Python environment locked
☐ SCRFD detection verified
☐ Alignment and embeddings verified
☐ Dynamic gallery demonstrated
☐ Research questions approved
☐ Permission and consent approved
☐ Pilot completed and reviewed
☐ Main dataset collected and backed up
☐ Metadata and CVAT verification complete
☐ Three identity-disjoint splits frozen
☐ ArcFace and AdaFace baselines complete
☐ Augmentation profiles validated
☐ Domain Adapter trained on development identities only
☐ Enrollment ablation complete
☐ Temporal aggregation ablation complete
☐ Open-set thresholds frozen from development data
☐ Three-split final evaluation complete
☐ Runtime and condition-wise results complete
☐ FaceLiVTv2 comparison complete or transparently omitted
☐ Confidence intervals and failure analysis complete
☐ Paper tables match raw results
☐ Dataset/model cards complete
☐ Clean-environment rerun successful
☐ Minimal demo complete after research
☐ Privacy review and final deletion/retention plan complete

Appendix A. Suggested Script Command Flow

```
# 1. Frame extraction
python scripts/extract_frames.py   --input-root <RAW_VIDEO_ROOT>   --output-root <FRAME_ROOT>   --sample-
fps 5

# 2. Detection and alignment
python scripts/detect_faces.py --config configs/models.yaml
python scripts/align_faces.py --config configs/models.yaml

# 3. Tracking and verification
python scripts/build_tracks.py --config configs/models.yaml

# 4. Embeddings
python scripts/extract_embeddings.py   --backbone adaface_ir50   --split development

# 5. Adapter data and training
python scripts/prepare_adapter_data.py --split development
python scripts/train_adapter.py --config configs/adapter.yaml

# 6. Gallery and threshold calibration
python scripts/build_gallery.py --split development
python scripts/calibrate_thresholds.py --config configs/evaluation.yaml

# 7. Final evaluation
python scripts/evaluate_open_set.py --split split_01 --frozen-threshold T01
python scripts/evaluate_open_set.py --split split_02 --frozen-threshold T01
python scripts/evaluate_open_set.py --split split_03 --frozen-threshold T01

# 8. Analysis and runtime
python scripts/evaluate_augmentations.py
python scripts/evaluate_aggregation.py
python scripts/benchmark_runtime.py
```

Appendix B. Example Result Schema

```
probe_track_id,subject_id,ground_truth_status,camera,session,
face_count,mean_face_width,track_quality,top1_id,top1_score,top2_score,
margin,decision,is_correct,backbone,adapter_version,enrollment_method,
aggregation_method,threshold_version,split

T0001,S031,KNOWN,CAMA,C2,24,38.5,0.72,S031,0.61,0.48,
0.13,KNOWN,1,AdaFace,ADP01,augmented_template,quality_weighted,T01,split_01
```

Appendix C. Reference and Source Register

References below are included to define model, dataset, tool, and metric choices. Access dates and repository versions should be recorded in the final paper and README.

ID	Source	Use in this project
[P0]	Existing FYDP report	A Real-Time CCTV-Based Face Detection and Recognition System for Educational Institutions Using Lightweight Deep Learning Models, July 11, 2026. User-provided source report.
[R1]	SCRFD	Sample and Computation Redistribution for Efficient Face Detection; InsightFace project and official SCRFD repository/documentation.
[R2]	ArcFace	Deng et al., ArcFace: Additive Angular Margin Loss for Deep Face Recognition, CVPR 2019.
[R3]	AdaFace	Kim, Jain, and Liu, AdaFace: Quality Adaptive Margin for Face Recognition, CVPR 2022; official repository.

ID	Source	Use in this project
[R4]	ByteTrack	Zhang et al., ByteTrack: Multi-Object Tracking by Associating Every Detection Box, ECCV 2022; official repository.
[R6]	QMUL-SurvFace	Cheng, Zhu, and Gong, Surveillance Face Recognition Challenge; official QMUL-SurvFace project page.
[R7]	WIDER FACE	Yang et al., WIDER FACE: A Face Detection Benchmark, CVPR 2016.
[R8]	SCface	Grgic, Delac, and Grgic, SCface - Surveillance Cameras Face Database.
[R9]	RFW	Wang et al., Racial Faces in-the-Wild, ICCV 2019; official testing page.
[R10]	FaceLiVTv2	Setyawan et al., FaceLiVTv2: An Improved Hybrid Architecture for Efficient Mobile Face Recognition, arXiv:2604.09127, 2026; official repository. Preprint status.
[R11]	Albumentations	Official Albumentations documentation for blur, compression, downscale, and other image transforms.
[R12]	CVAT	Official CVAT documentation for installation, video annotation, and Track Mode.
[R13]	FAISS	Meta AI Research, FAISS: efficient similarity search and clustering of dense vectors; official repository.
[R14]	IJB-C	Maze et al., IARPA Janus Benchmark-C: Face Dataset and Protocol, ICB 2018; NIST challenge page.
[R15]	NIST metrics	NIST Face Recognition Technology Evaluation 1:N and 1:1 pages defining FPIR, FNIR, FMR, and FNMR.
[R16]	PyTorch	Official PyTorch Get Started installation selector and documentation.

Appendix D. Official Web Resources

InsightFace / SCRFD: https://github.com/deepinsight/insightface
AdaFace official repository: https://github.com/mk-minchul/adaface
ArcFace project: https://www.insightface.ai/research/arcface
ByteTrack official repository: https://github.com/FoundationVision/ByteTrack
QMUL-SurvFace: https://qmul-survface.github.io/
SCface: https://www.scface.org/
WIDER FACE: https://shuoyang1213.me/WIDERFACE/
RFW testing: https://whdeng.cn/RFW/testing.html
FaceLiVT official repository: https://github.com/novendrastywn/FaceLiVT
CVAT documentation: https://docs.cvat.ai/
Albumentations documentation: https://albumentations.ai/docs/
FAISS official repository: https://github.com/facebookresearch/faiss
NIST FRTE 1:N: https://pages.nist.gov/frvt/html/frvt1N.html
PyTorch installation: https://pytorch.org/get-started/locally/

Final operational rule

Start with Week 1. Do not skip forward to the Domain Adapter, transformer comparison, or main dataset collection until the preceding gate is complete. The project becomes publishable through disciplined data, splits, calibration, ablations, and reproducibility - not through adding unrelated features.